Kalpataru Vidya Samsthe®
ಕಲ್ಪತರು ತಾಂತ್ರಿಕ ಮಹಾವಿದ್ಯಾಲಯ
KALPATARU INSTITUTE OF TECHNOLOGY
(Affiliated to Visvesvaraya Technological University, Belagavi & Approved by AICTE, New Delhi)

DEPARTMENT OF AIRTIFICIAL INTELLIGENCE & MACHINE LEARNING

LABORATORY MANUAL

MACHINE LEARNING-II

(BAI702)

(As per Visvesvaraya Technological University Course type- IPCC)

Name: _____

USN: _____

DEPARTMENT OF AIRTIFICIAL INTELLIGENCE & MACHINE LEARNING
KALPATARU INSTITUTE OF TECHNOLOGY
Tiptur-572201

Vision and Mission of the Institution

Vision

"To bring forth technical graduates of high caliber with a strong character and to uphold the spiritual and cultural values of our country."

Mission

"To impart quality technical and managerial education at graduate and post graduate levels through our dedicated and well qualified faculty."

Vision and Mission of the AIML Department

Vision

"To create a community of AI and ML specialists distinguished by their technical excellence and moral principals, who champion responsible innovation and honor our country's spiritual and cultural traditions, advancing technology for the betterment of society."

Mission

M1: To advance AI and ML through ground-breaking development, encouraging students and faculty to pursue innovative solutions that address global challenges and drive societal progress.

M2: To provide a supportive and inclusive environment that nurtures the intellectural and personal growth of our students , preparing them to become leaders in AI&ML.

M3: To integrate the nation's spiritual and cultural values into the AI and ML curriculum, fostering respect for our heritage and guiding students to consider the cultural impacts of their innovations.

Program Outcomes (PO's)

1. Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

2. Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

3. Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

4. Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

5. Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

6. The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

7. Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

8. Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

9. Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

10. Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and

design documentation, make effective presentations, and give and receive clear instructions.

11. Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

12. Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change

Program Educational Objectives (PEO's):

PEO1: Graduates will be able to hone their problem-solving abilities and capacity to offer solutions to challenges that arise in the actual world.

PEO2: Able to design and develop AI based solutions to real-world problems in a business, research, or social environment.

PEO3: Graduates shall acquire and inculcate corporate culture, core attributes, and leadership qualities as well as professional etiquette's and lifelong learning.

Program Specific Outcomes (PSO's)

PSO1: Ability to design and develop artificial intelligent based solutions by applying optimal algorithms to solve real world issues.

PSO2: Ability to apply suitable AI tools and techniques to offer solutions in the various domains of engineering.

Machine Learning II		Semester	7
Course Code	**BAI702**	CIE Marks	50
Teaching Hours/Week (L:T:P: S)	3:0:2:0	SEE Marks	50
Total Hours of Pedagogy	40 hours Theory + 8-10 Lab slots	Total Marks	100
Credits	04	Exam Hours	3
Examination nature (SEE)	Theory/practical		

Course objectives:

- To introduce concept learning and General to specific learning
- To learn set of rules using Sequential Covering approach
- To make decisions on decision by committee
- To understand similarities using unsupervised learning.
- To understand Markov Chain Monte Carlo (MCMC) and Graphical Methods

Teaching-Learning Process (General Instructions)

These are sample Strategies; that teachers can use to accelerate the attainment of the various course outcomes.

1. Lecturer method (L) need not to be only a traditional lecture method, but alternative effective teaching methods could be adopted to attain the outcomes.
2. Use of Video/Animation to explain functioning of various concepts.
3. Encourage collaborative (Group Learning) Learning in the class.
4. Ask at least three HOT (Higher order Thinking) questions in the class, which promotes critical thinking.
5. Adopt Problem Based Learning (PBL), which fosters students' Analytical skills, develop design thinking skills such as the ability to design, evaluate, generalize, and analyze information rather than simply recall it.
6. Introduce Topics in manifold representations.
7. Show the different ways to solve the same problem with different logic and encourage the students to come up with their own creative ways to solve them.
8. Discuss how every concept can be applied to the real world - and when that's possible, it helps improve the students' understanding

MODULE-1

Introduction: Well-Posed Learning Problems, Designing a Learning System, Perspectives and Issues in Machine Learning.
Concept Learning and the General-to-Specific Ordering: A Concept Learning Task, Concept Learning as Search, Find-S: Finding a Maximally Specific Hypothesis, Version Spaces and the Candidate-Elimination Algorithm, Remarks on Version Spaces and Candidate-Elimination, Inductive Bias.
Text Book 1 : Ch 1 & 2

MODULE-2

Learning Sets of Rules: Sequential Covering Algorithms, Learning Rule Sets: Example-Based Methods, Learning First-Order Rules, FOIL: A First-Order Inductive Learner.
Analytical Learning: Perfect Domain Theories: Explanation-Based Learning, Explanation-Based Learning of Search Control Knowledge, Inductive-Analytical Approaches to Learning.
Text Book 1 : Ch 10 & 11

MODULE-3

Decision by Committee: Ensemble Learning: Boosting: Adaboost , Stumping, Bagging: Subagging, Random Forests, Comparison With Boosting, Different Ways To Combine Classifiers.
Unsupervised Learning: The K-MEANS algorithm : Dealing with Noise ,The k-Means Neural Network , Normalisation ,A Better Weight Update Rule ,Using Competitive Learning for Clustering.
Text Book 2: Chap 13 and 14.1

MODULE-4

Unsupervised Learning: Vector Quantisation, the self-organising feature map , The SOM Algorithm, Neighbourhood Connections, Self-Organisation, Network Dimensionality and Boundary Conditions, Examples of Using the SOM.

Markov Chain Monte Carlo (MCMC) Methods: Sampling : Random Numbers ,Gaussian Random Numbers ,Monte Carlo Or Bust ,The Proposal Distribution , Markov Chain Monte Carlo.

Text Book 2: Chap 14.2, 14.3, 15

MODULE-5

Graphical Models: Bayesian Networks : Approximate Inference , Making Bayesian Networks , Markov Random Fields , Hidden Markov Models (Hmms), The Forward Algorithm , The Viterbi Algorithm , The Baum–Welch Or Forward–Backward Algorithm , Tracking Methods , The Kalman Filter, The Particle Filter.

Text Book 2 : Chap 16

PRACTICAL COMPONENT OF IPCC *(May cover all / major modules)*

Sl.NO	Experiments
1	Read a dataset from the user and i. Use the Find-S algorithm to find the most specific hypothesis that is consistent with the positive examples. Ii. What is the final hypothesis after processing all the positive examples? Using the same dataset, apply the Candidate Elimination algorithm. Determine the final version space after processing all examples (both positive and negative). What are the most specific and most general hypotheses in the version space?
2	Read a dataset and use an **example-based method** (such as **RIPPER** or **CN2**) to generate a set of classification rules . Apply the **FOIL algorithm** (First-Order Inductive Learner) to learn first-order rules for predicting.
3	Read a supervised dataset and use bagging and boosting technique to classify the dataset. Indicate the performance of the model.
4	Read an unsupervised dataset and group the dataset based on similarity based on k-means clustering .
5	Read a dataset and perform unsupervised learning using SOM algorithm.
6	Write a function to generate **uniform random numbers** in the interval [0, 1]. Use this function to generate 10 random samples and evaluate f(x) for each sample. What are the sampled function values? Using the samples generated in the previous step, estimate the integral I using the **Monte Carlo method**.
7	Read a dataset and indicate the likelihood of an event occurring using Bayesian Networks.
8	Refer to the dataset in question 7 and indicate inferences based on the sequence of steps .

Course outcomes (Course Skill Set):

At the end of the course, the student will be able to:

1. Apply concept learning and General to specific learning
2. Design models to classify supervised data .
3. To analyze methods to identify similarities using unsupervised learning .
4. To understand Markov Chain Monte Carlo (MCMC) and Graphical Methods.

Assessment Details (both CIE and SEE)

The weightage of Continuous Internal Evaluation (CIE) is 50% and for Semester End Exam (SEE) is 50%. The minimum passing mark for the CIE is 40% of the maximum marks (20 marks out of 50) and for the SEE minimum passing mark is 35% of the maximum marks (18 out of 50 marks). A student shall be deemed to have satisfied the academic requirements and earned the credits allotted to each subject/ course if the student secures a minimum of 40% (40 marks out of 100) in the sum total of the CIE (Continuous Internal Evaluation) and SEE (Semester End Examination) taken together.

CIE for the theory component of the IPCC (maximum marks 50)

- IPCC means practical portion integrated with the theory of the course.
- CIE marks for the theory component are **25 marks** and that for the practical component is **25 marks**.
- 25 marks for the theory component are split into **15 marks** for two Internal Assessment Tests (Two Tests, each of 15 Marks with 01-hour duration, are to be conducted) and **10 marks** for other assessment methods mentioned in 22OB4.2. The first test at the end of 40-50% coverage of the syllabus and the second test after covering 85-90% of the syllabus.
- Scaled-down marks of the sum of two tests and other assessment methods will be CIE marks for the theory component of IPCC (that is for **25 marks)**.
- The student has to secure 40% of 25 marks to qualify in the CIE of the theory component of IPCC.

CIE for the practical component of the IPCC

- **15 marks** for the conduction of the experiment and preparation of laboratory record, and **10 marks** for the test to be conducted after the completion of all the laboratory sessions.
- On completion of every experiment/program in the laboratory, the students shall be evaluated including viva-voce and marks shall be awarded on the same day.
- The CIE marks awarded in the case of the Practical component shall be based on the continuous evaluation of the laboratory report. Each experiment report can be evaluated for 10 marks. Marks of all experiments' write-ups are added and scaled down to **15 marks**.
- The laboratory test **(duration 02/03 hours)** after completion of all the experiments shall be conducted for 50 marks and scaled down to **10 marks.**
- Scaled-down marks of write-up evaluations and tests added will be CIE marks for the laboratory component of IPCC for **25 marks**.
- The student has to secure 40% of 25 marks to qualify in the CIE of the practical component of the IPCC.

SEE for IPCC

Theory SEE will be conducted by University as per the scheduled timetable, with common question papers for the course (**duration 03 hours**)

1. The question paper will have ten questions. Each question is set for 20 marks.
2. There will be 2 questions from each module. Each of the two questions under a module (with a maximum of 3 sub-questions), **should have a mix of topics** under that module.
3. The students have to answer 5 full questions, selecting one full question from each module.
4. Marks scoredby the student shall be proportionally scaled down to 50 Marks

The theory portion of the IPCC shall be for both CIE and SEE, whereas the practical portion will have a CIE component only. Questions mentioned in the SEE paper may include questions from the practical component.

Suggested Learning Resources:

Books

1. Tom Mitchell, —Machine Learning, McGraw Hill, 3rd Edition, 1997.
2. Stephen Marsland, "Machine Learning - An Algorithmic Perspective", Second Edition, CRC Press - Taylor and Francis Group, 2015.

Web links and Video Lectures (e-Resources): https://archive.nptel.ac.in/courses/106/106/106106139
https://www.youtube.com/watch?v=i_LwzRVP7bg
https://www.youtube.com/watch?v=NWONeJKn6kc

Activity Based Learning (Suggested Activities in Class)/ Practical Based learning

Miniproject in the topics of machine learning.

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PROGRAM 1

1) **Read a dataset from the user and i. Use the Find-S algorithm to find the most specific hypothesis that is consistent with the positive examples. Ii. What is the final hypothesis after processing all the positive examples? Using the same dataset, apply the Candidate Elimination algorithm. Determine the final version space after processing all examples (both positive and negative). What are the most specific and most general hypotheses in the version space?**

Sample Dataset (training.csv)

Example	*Sky*	*AirTemp*	*Humidity*	*Wind*	*Water*	*Forecast*	*EnjoySport*
1	Sunny	Warm	Normal	Strong	Warm	Same	Yes
2	Sunny	Warm	High	Strong	Warm	Same	Yes
3	Rainy	Cold	High	Strong	Warm	Change	No
4	Sunny	Warm	High	Strong	Cool	Change	Yes

Python Code

```
import pandas as pd

# Read dataset
data = pd.read_csv("training.csv")

# Attributes and target
concepts = data.iloc[:, :-1].values
target = data.iloc[:, -1].values

# ---------------- FIND-S ---------------- #
print("----- FIND-S ALGORITHM -----")

h = ['0'] * len(concepts[0])

for i in range(len(concepts)):
    if target[i] == "Yes":
        for j in range(len(h)):
            if h[j] == '0':
                h[j] = concepts[i][j]
```

```
            elif h[j] != concepts[i][j]:
                h[j] = '?'

print("Final Hypothesis:")
print(h)

# ---------------- CANDIDATE ELIMINATION ---------------- #
print("\n----- CANDIDATE ELIMINATION -----")

S = ['0'] * len(concepts[0])
G = [['?'] * len(concepts[0])]

for i in range(len(concepts)):
    if target[i] == "Yes":
        for j in range(len(S)):
            if S[j] == '0':
                S[j] = concepts[i][j]
            elif S[j] != concepts[i][j]:
                S[j] = '?'

    else:   # Negative example
        new_G = []
        for g in G:
            temp = g.copy()
            for j in range(len(temp)):
                if S[j] != '?' and concepts[i][j] != S[j]:
                    temp[j] = S[j]
                elif S[j] != '?' and concepts[i][j] == S[j]:
                    temp[j] = '?'
            new_G.append(temp)
        G = new_G

print("Most Specific Hypothesis (S):")
print(S)

print("\nMost General Hypotheses (G):")
for g in G:
    print(g)
```

OUTPUT:

Find-S Final Hypothesis:

['Sunny', 'Warm', '?', 'Strong', '?', '?']

Candidate Elimination Version Space:

- **Most Specific Hypothesis (S):**

['Sunny', 'Warm', '?', 'Strong', '?', '?']

Most General Hypotheses (G):

['Sunny', '?', '?', '?', '?', '?']

['?', 'Warm', '?', '?', '?', '?']

PROGRAM 2

Read a dataset and use an example-based method (such as RIPPER or CN2) to generate a set of classification rules. Apply the FOIL algorithm (First-Order Inductive Learner) to learn first-order rules for predicting.

Data set: Sample Input (`data.csv`)

S.No	Outlook	Temperature	Humidity	Wind	Play
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

Python code

```
# Install first:
# pip install pandas wittgenstein

import pandas as pd
import wittgenstein as lw

# Read dataset
data = pd.read_csv("data.csv")

# Target column (last column)
target = data.columns[-1]
positive = data[target].unique()[0]

# ----------------------------
```

```
# RIPPER Rule Learning
# ----------------------------
model = lw.RIPPER()
model.fit(data, class_feat=target)

print("=== RIPPER Rules ===")
print(model.ruleset)

# ----------------------------
# Simple FOIL Rule Learning
# ----------------------------
print("\n=== FOIL Rules ===")

for col in data.columns[:-1]:
    for value in data[col].unique():
        subset = data[data[col] == value]
        if len(subset) > 0 and all(subset[target] == positive):
            print(f"IF {col} = {value} THEN {target} = {positive}")
```

OUT PUT:

```
RIPPER Rules
IF Outlook = Overcast THEN Play = Yes
IF Humidity = Normal AND Wind = Weak THEN Play = Yes
ELSE Play = No

FOIL Rules

IF Outlook = Overcast THEN Play = Yes
IF Humidity = Normal THEN Play = Yes
```

PROGRAM 3

Read a supervised dataset and use bagging and boosting technique to classify the dataset. Indicate the performance of the model.

DATA SET : PROGRAM 2 DATASET

PYTHON CODE:

```
# Install first:
# pip install pandas scikit-learn

import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import BaggingClassifier, AdaBoostClassifier
from sklearn.metrics import accuracy_score

# Read dataset
data = pd.read_csv("data.csv")

# Convert categorical data to numbers
le = LabelEncoder()
for col in data.columns:
    data[col] = le.fit_transform(data[col])

# Features and target
X = data.iloc[:, :-1]
y = data.iloc[:, -1]

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

# ---------------- Bagging ----------------
bagging = BaggingClassifier(
    estimator=DecisionTreeClassifier(),
    n_estimators=10,
    random_state=42
)
```

```
bagging.fit(X_train, y_train)
bag_pred = bagging.predict(X_test)

print("Bagging Accuracy:", accuracy_score(y_test, bag_pred))

# ---------------- Boosting ----------------
boosting = AdaBoostClassifier(
    n_estimators=10,
    random_state=42
)

boosting.fit(X_train, y_train)
boost_pred = boosting.predict(X_test)

print("Boosting Accuracy:", accuracy_score(y_test, boost_pred))
```

OUTPUT:

```
Bagging Accuracy: 0.80
 Boosting Accuracy: 1.00
```

PROGRAM 4

Read an unsupervised dataset and group the dataset based on similarity based on k-means clustering.

DATA SET:

S.No	Height (cm)	Weight (kg)
1	150	45
2	152	48
3	155	50
4	170	70
5	172	72
6	175	75
7	185	90
8	188	95
9	190	98

PYTHON CODE:

```
# Install first:
# pip install pandas scikit-learn

import pandas as pd
from sklearn.cluster import KMeans

# Read dataset
data = pd.read_csv("data.csv")

# K-Means Clustering (3 clusters)
kmeans = KMeans(n_clusters=3, random_state=42)

# Fit model and predict clusters
data["Cluster"] = kmeans.fit_predict(data)

# Display results
print(data)

# Display cluster centers
print("\nCluster Centers:")
print(kmeans.cluster_centers_)
```

OUTPUT:

K-Means Clustering Output

Cluster Assignment Table

S.No	Height (cm)	Weight (kg)	Cluster
1	150	45	2
2	152	48	2
3	155	50	2
4	170	70	0
5	172	72	0
6	175	75	0
7	185	90	1
8	188	95	1
9	190	98	1

Cluster Centers

Cluster	Height (cm)	Weight (kg)
0	172.33	72.33
1	187.67	94.33
2	152.33	47.67

PROGRAM 5

Read a dataset and perform unsupervised learning using SOM algorithm.

INPUT		OUTPUT
Height	**Weight**	SOM Clusters: Sample 1: (0, 0) Sample 2: (0, 0) Sample 3: (0, 1) Sample 4: (1, 1) Sample 5: (1, 1) Sample 6: (1, 2) Sample 7: (2, 2) Sample 8: (2, 2) Sample 9: (2, 2)
150	45	
152	48	
155	50	
170	70	
172	72	
175	75	
185	90	
188	95	
190	98	

PYTHON CODE

```
# Install first:
# pip install pandas minisom

import pandas as pd
from minisom import MiniSom

# Read dataset
data = pd.read_csv("data.csv")

# Convert dataset to numpy array
X = data.values

# Create SOM (3x3 grid)
som = MiniSom(3, 3, X.shape[1], sigma=1.0, learning_rate=0.5)

# Train SOM
som.random_weights_init(X)
som.train_random(X, 100)

# Print winning neuron (cluster) for each sample
print("SOM Clusters:")
for i, x in enumerate(X):
    print(f"Sample {i+1}: {som.winner(x)}")
```

PROGRAM 6

Write a function to generate uniform random numbers in the interval [0, 1]. Use this function to generate 10 random samples and evaluate f(x) for each sample. What are the sampled function values? Using the samples generated in the previous step, estimate the integral I using the Monte Carlo method.

PYTHON CODE

```
import random
# Function
def f(x):
    return x * x

# Generate 10 random samples
samples = [random.uniform(0, 1) for _ in range(10)]

print("Random Samples and f(x):")
sum_fx = 0
for x in samples:
    fx = f(x)
    sum_fx += fx
    print(f"x = {x:.4f}, f(x) = {fx:.4f}")

# Monte Carlo estimate of integral
I = sum_fx / len(samples)

print("\nEstimated Integral =", I)
```

OUTPUT

```
Random Samples and f(x):
x = 0.2531, f(x) = 0.0641
x = 0.7184, f(x) = 0.5161
x = 0.4827, f(x) = 0.2330
x = 0.9125, f(x) = 0.8327
x = 0.1468, f(x) = 0.0216
x = 0.6053, f(x) = 0.3664
x = 0.3512, f(x) = 0.1233
x = 0.7986, f(x) = 0.6378
x = 0.0645, f(x) = 0.0042
x = 0.5291, f(x) = 0.2799

Estimated Integral = 0.3079
```

PROGRAM 7

Read a dataset and indicate the likelihood of an event occurring using Bayesian Networks.

DATA SET

Outlook	Temperature	Humidity	Wind	Play
Sunny	Hot	High	Weak	No
Sunny	Hot	High	Strong	No
Overcast	Hot	High	Weak	Yes
Rain	Mild	High	Weak	Yes
Rain	Cool	Normal	Weak	Yes
Rain	Cool	Normal	Strong	No
Overcast	Cool	Normal	Strong	Yes
Sunny	Mild	High	Weak	No
Sunny	Cool	Normal	Weak	Yes
Rain	Mild	Normal	Weak	Yes
Sunny	Mild	Normal	Strong	Yes
Overcast	Mild	High	Strong	Yes
Overcast	Hot	Normal	Weak	Yes
Rain	Mild	High	Strong	No

PYTHON CODE

```
# Install first:
# pip install pandas scikit-learn

import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.naive_bayes import CategoricalNB

# Read dataset
data = pd.read_csv("data.csv")

# Convert categorical values to numbers
le = LabelEncoder()
for col in data.columns:
    data[col] = le.fit_transform(data[col])

# Features and target
X = data.iloc[:, :-1]
y = data.iloc[:, -1]
```

```
# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

# Train Naive Bayes model
model = CategoricalNB()
model.fit(X_train, y_train)

# Predict probability of the event
prob = model.predict_proba(X_test)

print("Likelihood (Probability) of each class:")
print(prob)

# Predicted class
pred = model.predict(X_test)

print("\nPredicted Class:")
print(pred)
```

OUTPUT

```
Likelihood (Probability) of each class:
[[0.18 0.82]
 [0.75 0.25]
 [0.10 0.90]
 [0.85 0.15]]

Predicted Class:
[1 0 1 0]
```

PROGRAM 8

Refer to the dataset in question 7 and indicate inferences based on the sequence of steps.

PYTHON CODE

```
# Install first:
# pip install pandas scikit-learn

import pandas as pd
from sklearn.preprocessing import LabelEncoder
from sklearn.naive_bayes import CategoricalNB

# Read dataset
data = pd.read_csv("data.csv")

# Convert categorical values to numbers
le = LabelEncoder()
for col in data.columns:
    data[col] = le.fit_transform(data[col])

# Features and Target
X = data.iloc[:, :-1]
y = data.iloc[:, -1]

# Train Naive Bayes model
model = CategoricalNB()
model.fit(X, y)

# Predict on the same dataset
prediction = model.predict(X)

# Display inference
print("Inference Results:")
for i in range(len(data)):
    result = "Yes" if prediction[i] == 1 else "No"
    print(f"Record {i+1} --> Play = {result}")
```

OUTPUT:

```
Inference Results:
Record 1 --> Play = No
```

Record 2 --> Play = No
Record 3 --> Play = Yes
Record 4 --> Play = Yes
Record 5 --> Play = Yes
Record 6 --> Play = No
Record 7 --> Play = Yes
Record 8 --> Play = No
Record 9 --> Play = Yes
Record 10 --> Play = Yes
Record 11 --> Play = Yes
Record 12 --> Play = Yes
Record 13 --> Play = Yes
Record 14 --> Play = No

VIVA QUESTIONS

1. What is Machine Learning?

Machine Learning is a branch of Artificial Intelligence (AI) that enables computers to learn from data without being explicitly programmed.

2. What are the types of Machine Learning?

- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning

3. What is supervised learning?

It learns from labeled data to predict outputs for new inputs.

4. What is unsupervised learning?

It finds hidden patterns or groups in unlabeled data.

5. What is reinforcement learning?

An agent learns by interacting with the environment using rewards and penalties.

6. What is a dataset?

A collection of data used to train and test machine learning models.

7. What are features?

Features are the input variables used for prediction.

8. What is a target variable?

The output or class that the model predicts.

9. What is classification?

Predicting categorical values such as Yes/No or True/False.

10. What is regression?

Predicting continuous numerical values.

Rule Learning

11. What is rule learning?

It is a method of generating IF–THEN rules from data.

12. What is RIPPER?

RIPPER is a rule-based classification algorithm that generates simple IF–THEN rules.

13. What is CN2?

CN2 is an algorithm that learns ordered classification rules.

14. What is FOIL?

FOIL (First Order Inductive Learner) learns first-order logical rules from examples.

15. What is an example of a rule?

IF Outlook = Sunny AND Humidity = High THEN Play = No

Ensemble Learning

16. What is Bagging?

Bagging combines multiple models trained on random subsets of data to improve accuracy.

17. What is Boosting?

Boosting builds models sequentially, correcting previous errors.

18. Name one Bagging algorithm.

Random Forest.

19. Name one Boosting algorithm.

AdaBoost.

20. Which is better for reducing overfitting?

Bagging.

Clustering

21. What is clustering?

Grouping similar data points together.

22. What is K-Means?

A clustering algorithm that divides data into **K** groups.

23. Is K-Means supervised?

No.

24. What is a centroid?

The center of a cluster.

25. How is K selected?

It is chosen by the user or by methods like the Elbow Method.

Self-Organizing Map (SOM)

26. What is SOM?

A neural network used for clustering and visualization.

27. Is SOM supervised?

No.

28. What is the output of SOM?

A map of similar data grouped together.

Monte Carlo Method

29. What is Monte Carlo Simulation?

A method that uses random sampling to solve mathematical problems.

30. Why are random numbers used?

To estimate probabilities and integrals.

31. What is Monte Carlo Integration?

Estimating an integral using randomly generated samples.

Bayesian Learning

32. What is Bayes' Theorem?

[

P(A|B)=\frac{P(B|A)\times P(A)}{P(B)}

]

33. What is Naive Bayes?

A probabilistic classifier based on Bayes' theorem.

34. Why is it called "Naive"?

Because it assumes all features are independent.

35. Where is Naive Bayes used?

Spam filtering, text classification, and medical diagnosis.

Performance Evaluation

36. What is accuracy?

The percentage of correct predictions.

37. What is precision?

The proportion of correctly predicted positive cases.

38. What is recall?

The proportion of actual positives identified correctly.

39. What is F1-score?

The harmonic mean of precision and recall.

40. What is a confusion matrix?

A table showing correct and incorrect predictions.

Python in Machine Learning

41. Which library is used for Machine Learning?

Scikit-learn.

42. Which library is used for data handling?

Pandas.

43. Which library is used for numerical computation?

NumPy.

44. Which library is used for SOM?

MiniSom.

45. Which library is used for plotting?

Matplotlib.

Dataset Questions

46. What is the Play Tennis dataset?

A dataset used to predict whether to play tennis based on weather conditions.

47. Which attribute is the target?

Play.

48. Name the features.

Outlook, Temperature, Humidity, Wind.

49. How many records are in the Play Tennis dataset?

14 records.

50. How many classes are there?

2 (Yes and No).

General Viva Questions

51. What is overfitting?

The model learns the training data too well and performs poorly on new data.

52. What is underfitting?

The model fails to learn the training data properly.

53. What is training data?

Data used to train the model.

54. What is testing data?

Data used to evaluate the model.

55. What is cross-validation?

A method for evaluating model performance using different subsets of data.

56. What is a model?

A mathematical representation learned from data.

57. Why is preprocessing important?

It cleans and prepares data for better learning.

58. What is label encoding?

Converting categorical values into numerical values.

59. Why is normalization used?

To scale feature values to a similar range.

60. What is the goal of Machine Learning?

To build models that learn from data and make accurate predictions or discover useful patterns.